10th GRADE BIOLOGY · TERM 1 · 1st COMMON WRITTEN EXAM

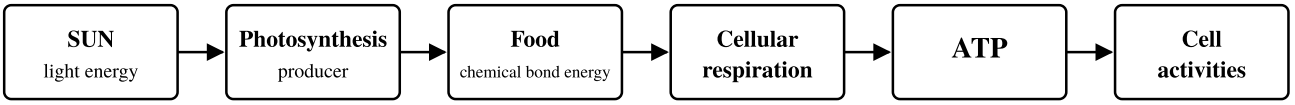
STUDY GUIDE — SENARYO 2

Your exam has 6 questions: ① Energy and ATP (BİY.10.1.1) → **3 Q** · ② Photosynthesis experiments (BİY.10.1.3) → **1 Q** · ③ Types and structures of digestion (BİY.10.1.5) → **2 Q**. Half the exam is ATP, so spend most of your time there.
The paper is in Turkish, so the Turkish exam words are given in [brackets]. Sources: your class notes (Energy, Photosynthesis, poster) and the MEB "Yazılıya Hazırlanıyorum" Biyoloji 10 sample questions.

1. ENERGY AND ATP FOR LIFE [canlılık için enerji ve ATP] (3 Q)

1.1 Energy and energy flow [enerji akışı]

Energy is the capacity to do work or cause a change. It comes in forms such as light, heat, electrical, sound, motion (potential, kinetic), chemical and nuclear energy. Energy cannot be created or destroyed; it only changes from one form to another. The main energy source on Earth is the **Sun**.

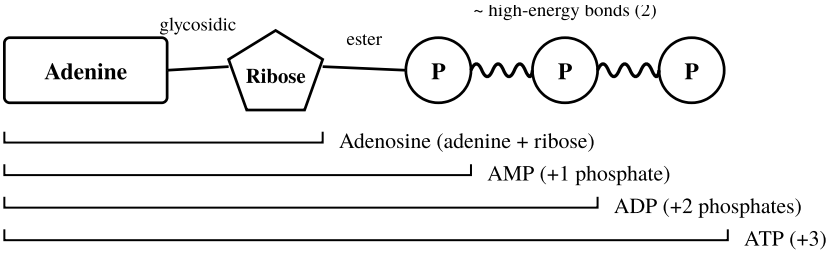


At every step part of the energy is lost as heat.

- **Autotrophs [ototrof]** make their own food (photosynthesizers: plants, algae, euglena [öglena], cyanobacteria, purple sulfur bacteria). **Heterotrophs [heterotrof]** eat ready-made food.
- Example food chain: Sun → grass → gazelle → lion. The energy in a lion's muscle cell also comes from the Sun: grass stores light energy in food, that energy passes along the food chain to the lion, and the lion transfers it to ATP by respiration.

1.2 Structure of ATP [ATP'nin yapısı]

ATP (adenosine triphosphate) [adenozin trifosfat] is the common energy molecule that every living thing uses **directly** for its life activities.



- Building blocks: **adenine** (nitrogenous organic base [azotlu organik baz]) + **ribose** (5-carbon sugar [5 karbonlu şeker]) + **3 phosphates** [fosfat]. There are **2 high-energy bonds** [yüksek enerjili bağ] between the phosphates.
- ATP is **a nucleotide but not a nucleic acid** [nükleotit yapılı, nükleik asit değil]. It looks like the adenine nucleotide of RNA (adenine + ribose); the difference is that it carries 3 phosphates instead of 1.

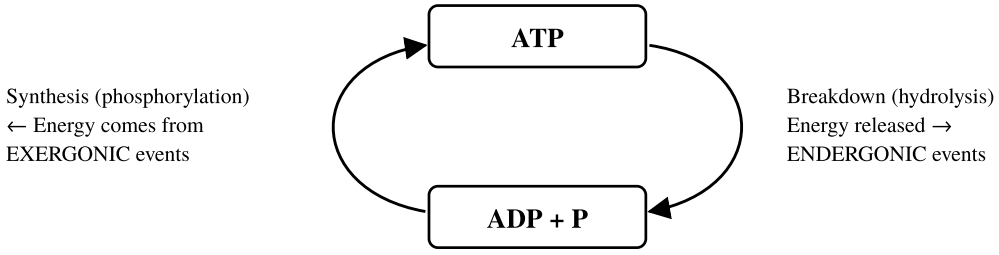
1.3 Properties of ATP — why ATP?

Property	What it means / how the exam asks it
Cannot cross the cell membrane and is not transported between cells	"A muscle cell sends its ATP to a nerve cell through the blood" → **false**; every cell makes its own ATP
Is not stored; made and used immediately	It is unstable and its bonds break easily. Long-term storage: glycogen, starch, fat
Releases energy gradually, in small packets [kademeli]	If glucose were broken down in one go, most of the energy would be lost as heat and the cell would be damaged
Common to all living things and used fast	"Universal energy molecule"; its energy can be converted to electrical, heat, motion and other forms

1.4 Making and breaking ATP — the ATP cycle [ATP döngüsü]

ADP + P + energy ⇄ ATP + H_2O

Direction	Name	Reaction type	Water	Energy
ADP + P → ATP	Phosphorylation [fosforilasyon] (ATP synthesis)	Dehydration [dehidrasyon]	Released	Used (stored)
ATP → ADP + P	Dephosphorylation [defosforilasyon] (ATP breakdown)	Hydrolysis [hidroliz]	**Used up**	Released



- **Endergonic (energy-absorbing) [endergonik] events:** protein and DNA synthesis (biosynthesis), active transport, muscle contraction, nerve impulse conduction [sinirsel iletim], cell division, endocytosis/exocytosis.
- **Exergonic (energy-releasing) [ekzergonik] events:** aerobic respiration [oksijenli solunum], fermentation [fermantasyon].
- ATP carries energy between these two kinds of events. Because it is constantly broken down and rebuilt from ADP + P, it is called a **renewable** [yenilenebilir] molecule.

1.5 Events that use ATP and events that don't

ATP is used	ATP is NOT used
Active transport, endocytosis, exocytosis, muscle contraction, protein/glycogen synthesis, cell division, nerve impulse conduction, mechanical digestion (muscle movement)	Diffusion, facilitated diffusion, osmosis (passive transport: high → low concentration on its own), hydrolysis reactions (chemical digestion), O_2–CO_2 exchange in the alveoli

EXTRA (NOT IN YOUR NOTES — ONLY IF YOUR TEACHER COVERED IT): ways ATP is made: **substrate-level phosphorylation** [substrat düzeyinde fosforilasyon] (cytoplasm and mitochondria, e.g. glycolysis), **oxidative phosphorylation** [oksidatif fosforilasyon] (mitochondria), **photophosphorylation** [fotofosforilasyon] (chloroplast, with light). Mature red blood cells have no mitochondria, so they make ATP only at the substrate level.

1.6 MEB-style experiment questions (this is how the exam asks)

- **Yeast experiment (MEB sample):** dish 1: glucose + yeast → yeast multiplies. Dish 2: glucose + yeast + *a substance that blocks ATP production* → no growth. **Conclusion:** without ATP the cell cannot divide, do active transport or synthesis → it dies; energy is **essential** for life.
- **Muscle fibre experiment:** fibres do not contract with glucose but contract with ATP → the molecule used **directly** in energy-requiring events is ATP. In a living cell, glucose is first turned into ATP by respiration.
- **Variables:** independent [bağımsız] = what the researcher changes (e.g. the added substance); dependent [bağımlı] = the measured result (e.g. growth, contraction); controlled [kontrol] = what is kept the same (temperature, cell count, time). Control group = the group without the test substance.

2. PHOTOSYNTHESIS EXPERIMENTS [fotosentez deneyleri] (1 Q)

$6CO_2 + 6H_2O \xrightarrow{\text{(light, chlorophyll)}} C_6H_{12}O_6 + 6O_2$

Used (inputs)	Produced (outputs)
Carbon dioxide (from air) · water (from soil) · light energy (from the Sun) · chlorophyll [klorofil] (absorbs light)	Glucose (organic food; stored in the leaf as **starch** [nişasta]) · oxygen (released to the air)

- The **O_2 released comes from water** (photolysis of water [suyun fotolizi]). The carbon and oxygen in glucose come from CO_2; the hydrogen comes from water.
- Visible light is 380–750 nm. Chlorophyll absorbs **violet-blue and red** light the most and **green the least** (it reflects green, so plants look green). Chlorophyll contains **Mg**; **Fe** is needed to make it. Accessory pigments [yardımcı pigmentler] absorb other wavelengths, pass the energy to chlorophyll and protect it.

Experiments you must know

Experiment	What is done / observed	Conclusion
Van Helmont	A willow sapling was grown for 5 years with only water; the tree gained ~74 kg, the soil lost only ~57 g	He said "the mass comes from water" — he missed the role of CO_2 from the air
Priestley	In a sealed jar a candle goes out and a mouse dies; with a mint plant in the jar the candle burns and the mouse lives	The plant "restores" the air → it produces O_2
Ingenhousz	Water plants release gas bubbles (O_2) in light, not in the dark; in the dark they give off CO_2	Photosynthesis needs light; only the green parts do it
Engelmann	Filamentous alga + prism + oxygen-loving (aerobic) bacteria: bacteria gather where violet-blue and red light fall	Photosynthesis is highest in violet-blue and red, lowest in green
Variegated leaf + iodine [alacalı yaprak]	Plant kept in the dark 48 h (old starch used up) → part of a leaf covered with foil → light → chlorophyll removed in alcohol → iodine. Green + lit part turns **blue-black**; white part and foil-covered part don't	Photosynthesis needs **chlorophyll** and **light**; the product is turned into starch
Elodea [su veba otu]	Inverted funnel + test tube; more bubbles as the lamp gets closer (MEB: violet light 10 cm → 120, 15 cm → 54, 20 cm → 30 bubbles / 2 min); no bubbles in the dark	Higher light intensity → faster photosynthesis; the gas is O_2
BTB (bromothymol blue)	Turns **yellow** when CO_2 rises, **blue** when CO_2 falls. Plant in light → blue; plant in the dark → yellow (respiration)	CO_2 is used in photosynthesis

- **Tests:** O_2 → re-ignites a glowing splint [kor hâlindeki kıymık] · starch → iodine turns blue-black · CO_2 → BTB turns yellow / limewater [kireç suyu] turns cloudy.
- **Factors affecting photosynthesis rate:** light intensity, light colour (wavelength), CO_2 amount, temperature. If increasing one factor no longer raises the rate, another factor is **limiting** [sınırlayıcı].
- If the lamp heats the water the result is invalid (temperature changes too) → put a water-filled container between the beaker and the lamp, or use an LED.

3. TYPES AND STRUCTURES OF DIGESTION [sindirim çeşitleri ve yapıları] (2 Q)

Digestion [sindirim] is breaking large food molecules into small building blocks that can pass through the cell membrane. **Mechanical digestion** [mekanik] is physical breakdown (chewing, grinding; muscles work, so ATP is used). **Chemical digestion** [kimyasal] is hydrolysis by enzymes (no ATP used).

	Intracellular digestion [hücre içi]	Extracellular digestion [hücre dışı]
How?	Food is taken in by phagocytosis → the food vacuole [besin kofulu] fuses with a **lysosome** [lizozom] → lysosome enzymes digest it	Enzymes are secreted outside the cell → food is digested outside (in the surroundings / digestive cavity / tract) → small molecules are absorbed
Examples	Amoeba, paramecium, human white blood cell [akyuvar] (engulfs bacteria)	Fungi, decomposer bacteria, carnivorous plants, earthworm, birds, mammals, humans
Both	Hydra, planarian [planarya]: first extracellular in the digestive cavity, then intracellular in the cells lining the cavity	

Lysosome enzymes are made on **ribosomes** (on rough ER) and packaged in the Golgi.

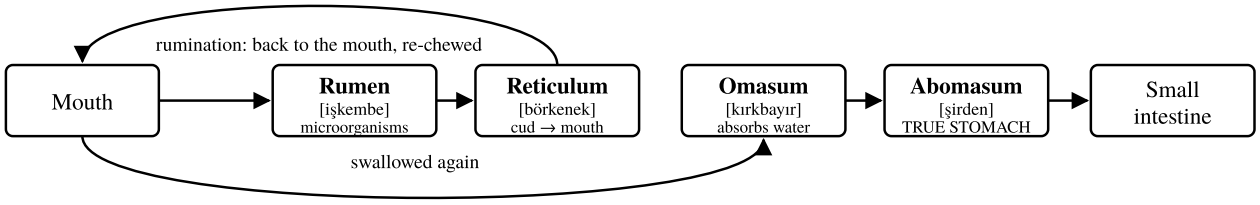
Digestive structures in invertebrates [omurgasızlar]

Animal	Structure	Note
Hydra	**One opening** [tek açıklıklı] (mouth only), sac-shaped gastrovascular cavity; tentacles with stinging cells around the mouth	Extracellular + intracellular
Planarian	**One opening**, branched digestive cavity; pharynx [yutak] that can extend out of the body	Extracellular + intracellular; food and waste use the same mouth
Earthworm [toprak solucanı]	**Two openings** [iki açıklıklı]: mouth → pharynx → oesophagus → **crop** [kursak] (stores, softens) → **gizzard** [taşlık] (muscular wall + sand grinds food) → intestine (chemical digestion, absorption; the typhlosole [tiflozol] increases surface area) → anus	One-way flow, specialised parts

Advantage of a two-opening system: food moves in one direction, each part specialises in a different job, and new food can be taken in while digestion continues → food is used more efficiently.

Digestive structures in vertebrates [omurgalılar]

- **Birds (no teeth):** crop [kursak] (stores and softens food) → glandular stomach [bezli mide] (secretes enzymes, chemical digestion) → **gizzard** [taşlık] (grinds with a muscular wall and swallowed small stones = does the job of teeth).
- **Ruminants [geviş getirenler] (cow, sheep):** 4-chambered stomach.



- The **abomasum** [şirden] is the true stomach: digestive enzymes are secreted here (it matches the human stomach).
- A cow cannot make cellulase [selülaz]; **bacteria and protists in the rumen** make cellulase and digest cellulose, and the cow uses the products and the microorganisms themselves (mutualism [karşılıklı yarar]).

Carnivore – herbivore – omnivore comparison (the MEB sample question asked this)

Criterion	Carnivore [etçil] (tiger, dog)	Herbivore [otçul] (cow, sheep)	Omnivore [hepçil] (human, bear)
Teeth	Sharp incisors, pointed well-developed canines → catch prey, tear meat	Wide, flat molars (grinding teeth) [azı dişleri]; canines missing or reduced → grind plants	Incisors, canines and molars all balanced
Stomach	One chamber	4 chambers (ruminants)	One chamber
Intestine	Short → meat is digested fast	Very long → fibrous, cellulose-rich food digests slowly and needs a longer digestion and absorption path	Medium length

EXAM TIP: if asked to "choose criteria and classify" [ölçütleri belirleyip sınıflandırınız], name all three criteria: **teeth, stomach (number of chambers), intestine length**. Then group each animal by those criteria with a reason and link it to its diet. **Write your answers in Turkish on the real exam.**